

Project Proposal: Parallel Online Examination System

1. Project Title

Parallel Online Examination System

2. Group Members

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3. Project Overview

A system where multiple students attempt exams at the same time while evaluator threads check their submissions in parallel; synchronization ensures correct result handling.

4. Features

Mandatory Features

- Student submission threads
- Evaluator threads
- Mutex-protected results
- Semaphore-controlled grading

Additional Features (if implemented)

- Time limits
- Auto-save
- Cheating detection
- Logs
- Analytics

5. Methodology

- Language: C
- Interface: CLI (GUI will be implemented only if possible)

- Use of threads for students and evaluators
- Mutex for protecting result storage
- Semaphore to limit grading capacity

6. Working

- Students submit exams using threads
- Evaluators check submissions in parallel
- Mutex ensures safe result storage
- Semaphore controls number of evaluators working at a time

7. Final Product

Exam dashboard with live status (CLI-based) which imitates a LMS system with Proper Synchronization.